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Effect of low-level laser therapy on postoperative pain after single-visit root canal retreatment of mandibular molars: A randomized controlled clinical trial

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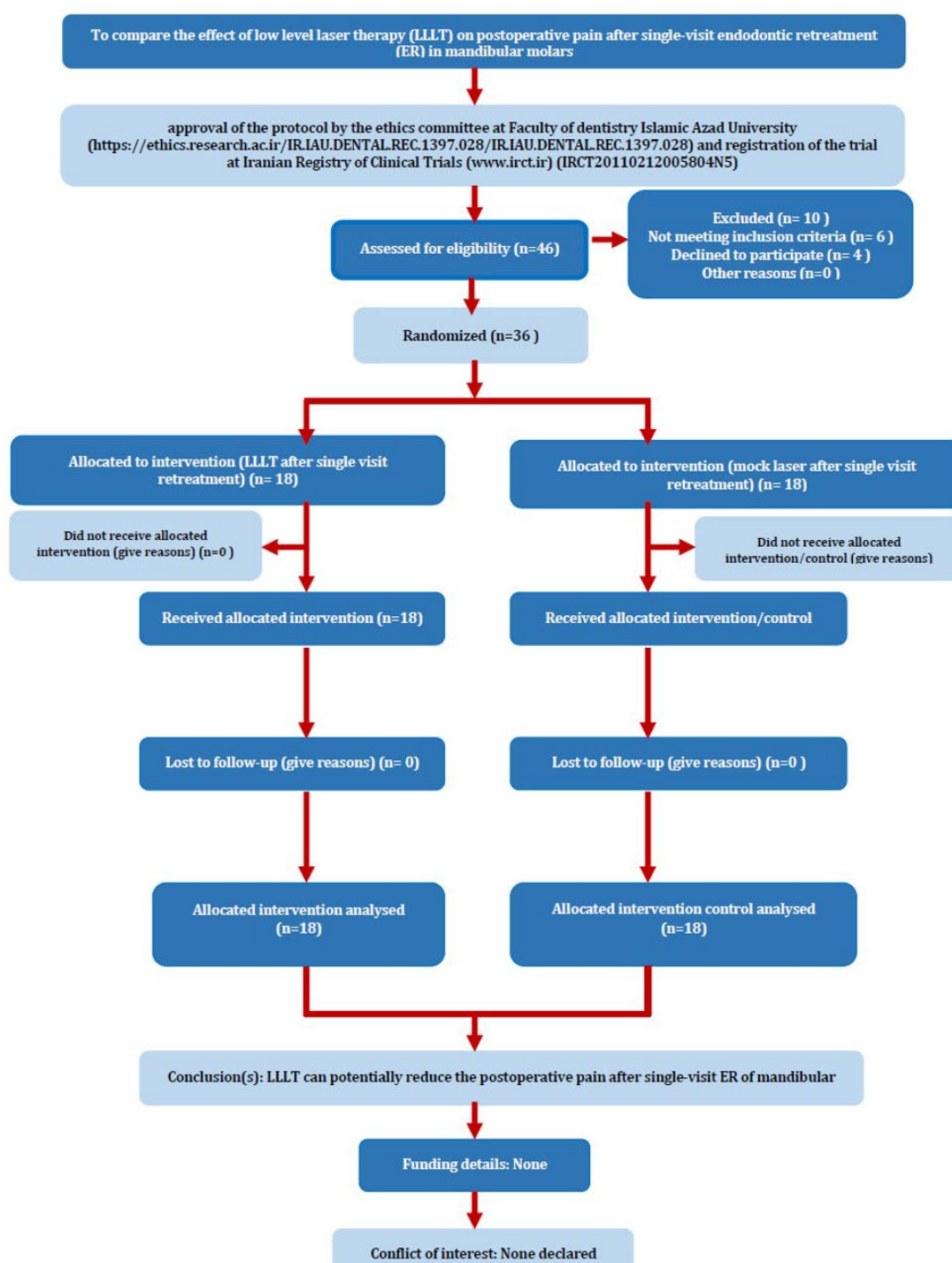
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Abstract

et al., 2020). In dentistry, the history of applying LLLT goes back to pain relief in orthodontic patients undergoing fixed appliance therapy (Turhani et al., 2006) and since then has been used as a tool for analgesia,

This randomized clinical trial has been written according to Preferred Reporting Items for Randomized Trials in Endodontics (PRIRATE) 2020 guidelines (Nagendrababu et al., 2020). Figure 1 sets out the key stages of the study.

PRIRATE 2020 Flowchart

*From: Nagendrababu V, Duncan HF, Bjørndal L, Kvist T, Priya E, Jayaraman J, Pulikkotil SJ, Pigg M, Rechenberg DK, Vaeth M, Dummer P. (2020) PRIRATE 2020 guidelines for reporting randomized trials in Endodontics: a consensus-based development. International Endodontic Journal Mar 20. doi: 10.1111/iej.13294. For further details, visit: <http://pride-endodonticguidelines.org/prirate/>

FIGURE 1 PRIRATE 2020 flowchart

Study design

This randomized, parallel-group sham-controlled trial was designed with two arms.

Subject enrolment

After approval of the protocol by the Ethics Committee at the Faculty of Dentistry, Islamic Azad University

(<https://ethics.research.ac.ir/IR.IAU.DENTAL.REC.1397.028>) and registration of the trial at Iranian Registry of Clinical Trials (www.irct.ir) (IRCT20110212005804N5) in December 2018, subject enrolment initiated from the pool of subjects requiring root canal retreatment of mandibular first and second molars who were referred to the Department of Endodontics, Faculty of Dentistry, Tehran Medical Sciences, Islamic Azad University, Iran, from March 2019 to March 2020. A fully informed consent was obtained from all approved patients.

Inclusion/exclusion criteria

Patients included in this study were in good health as determined by their medical history and verbal questioning and were scheduled for root canal retreatment on *symptomatic* first or second mandibular molars. Parallel periapical radiographies were taken, and mandibular molars with previous root canal treatment with a PAI index score of 2 or 3 (indicating small changes in bone structure and change in bone structure with mineral loss, respectively) according to the classification introduced by Ørstavik et al. (1986), were included. All patients were asked to report preoperative tooth and percussion pain on a 170-mm non-graded visual analogue scale (VAS), and those individuals with a spontaneous pain score of <56 mm on VAS and percussion pain of <56 mm were included. Patients were excluded in case of periodontal probing greater than 3 mm, swelling or sinus tract and palpation sensitivity.

Sample size calculation

The minimum sample size was calculated to be 18 in each of the study arms using the two-sample *t*-test feature of PASS 11 software assuming $\alpha = 0.05$ and $\beta = 0.2$, on the basis of the data obtained from the study by Arslan et al. (2017).

Randomization and concealment

Randomization was achieved using a Web-based generator (www.random.org) by a researcher (S. E.). The clinician who performed the retreatment procedure (E. E.) was blind to the group assignment. Another clinician performed laser irradiation in the test and sham groups (M. F.). Post-treatment patient follow-up was undertaken by another researcher (A. A.), and data analysis and interpretation were performed by another researcher (A. A. M.)

who was blind to the entire assignment and treatment procedure.

Root canal retreatment procedures

Patients were scheduled for a single-visit root canal retreatment. The patients' demographic data including age, gender and tooth number were recorded by the same clinician (S. E.). Patients were provided with two non-numeral 170-mm VAS to record their tooth pain and percussion pain, separately (as mentioned earlier, those with pain scores of less than 56 mm were included). Their preoperative pain values were recorded.

The root canal procedures were performed by one calibrated clinician (S. E.) following a standardized protocol. The procedure started with local anaesthesia using an inferior alveolar nerve block and injection of 1.8 mL 4% articaine with 1:200 000 epinephrine (Dentacain; Exir, Tehran, Iran) followed by rubber dam isolation. After access cavity preparation, the D RaCe retreatment system (FKG Dentaire, La Chaux-de-Fonds, Switzerland) was used with an X-Smart rotary motor (FKG Dentaire), as recommended by the manufacturer. After using DR1 size 30, 0.10 taper and DR2 size 25, 0.04 taper to remove most of the existing root-filling material and after preparing the coronal third of the canal, an attempt was made to achieve the estimated working length using a size 15 K-file (Mani, Tochigi, Japan). The canals were negotiated to the apical foramen, and when a canal obstruction prevented canal patency, the tooth was excluded. No chemical solvent was used to remove root filling.

The working length was determined 0.5 mm short of the apical foramen by means of an electronic apex locator (Root ZX; J. Morita USA, Austin, CA, USA) and then a periapical radiograph was taken to confirm the working length. After obtaining apical patency, a size 30 H-file (Mani) was used with a filing motion to remove remnants of the root filling. Root canal preparation was continued until no gutta-percha was observed on the instrument. Then more canal enlargement was done using two more complementary RaCe rotary instruments size 35, 0.04 taper for mesial canals, and size 40, 0.04 taper for distal canals (RaCe; FKG Dentaire). The canals were rinsed with 2 mL of 5.25% sodium hypochlorite with a 23-gauge syringe between instruments.

For smear layer removal, the final irrigation was performed with 5 mL of 5.25% sodium hypochlorite for 1 min followed by 5 mL of distilled water and then 2 mL of 17% EDTA (Asia-Shimi-Teb, Tehran, Iran) for 1 min. Then the root canals were dried with paper points (META Biomed Co. Ltd, Chungbuk, Korea) and then filled using the lateral compaction of master and accessory gutta-percha cones

(META Biomed Co. Ltd) and AH-Plus sealer (Dentsply, Tulsa Dental, Tulsa, OK, USA).

The access cavity was temporary filled with Cavit (3 M ESPE, St Paul, MN, USA), and a final radiograph was taken to confirm the quality of the root filling. Then the patients were randomly allocated for real or sham laser irradiation. The randomization procedure was achieved using a Web-based algorithm (www.random.org).

The LLLT and sham groups

Low-level laser therapy was administered by another clinician (M. F.). In the LLLT group, laser irradiation was performed using a 980-nm diode laser (Doctor Smile; Wiser, Lambda, Italy) set at 6.89 W/cm² energy density, 0.5 W power and a tip diameter of 10 mm for 15 s. The laser was irradiated to the soft tissues covering the mesial and distal apices of the tooth from the buccal side with the tip of the laser handpiece held at a distance of approximately 10 mm from the mucosa. In the sham group, the laser handpiece was positioned similar to that in the LLLT, group but the laser was not activated.

Post-treatment pain

All post-treatment follow-up sessions were carried out by the same clinician responsible for recording the patients' demographic data (S.E.) under the supervision of an endodontist (M. F.). After laser/sham treatment, each patient was given a home questionnaire and a second appointment was scheduled for permanent tooth restoration one week later. Each patient received 10 tablets of 500 mg acetaminophen (AlborzDarouCo Production, Qazvin, Iran) and were instructed to take them in case of pain. The patients were instructed to record their postoperative pain levels at 4, 8, 12 and 24 h and 2, 3 and 7 days after treatment on separate VAS scales. The patients were contacted by phone on specific hours and days to remind them to record their pain level. They were also asked to record consumption of analgesic on the questionnaire (dosage and exact time of taking the analgesic). If patients reported taking an analgesic, they were excluded from the study and replaced. Patients were free to ask for an unscheduled appointment, if needed, which was also recorded.

At the second visit scheduled 1 week later, the patients returned the questionnaires, and their chair side level of pain on palpation and percussion was recorded. Also, unexpected and unscheduled visits were recorded. The patients were then referred for permanent restoration of

the respective tooth by another clinician (A.A.). The intensity of pain experienced by patients was recorded using a graded VAS.

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